- It can serve as a base of attachment for the cytoskeleton in some organisms and the cell wall in others. Thus, the cell membrane also serves to maintain its shape.
- It can regulate the cell growth through the balance of endocytosis and exocytosis.
- It can maintain the concentration of water, inorganic ions and organic molecules between the cell and the environment.
- The plasma membrane also receives signals and coordinates molecular interactions at the surface such as cell to cell recognition, adhesion and communication.

1.4.2.Cell wall

The cell wall is a non-living rigid structure that forms an outer covering for the plasma membrane of fungi and plants. Cell wall not only gives shape to the cell and protects the cell from mechanical damage and infection, it also helps in cell-to-cell interaction and provides a barrier to undesirable macromolecules.

Bacterial cell wall:

Bacteria have a cell wall, which is a rigid, carbohydrate-containing structure that surrounds the bacterial cell. However, the genus Mycoplasma, do not have cell wall. The cell wall provides the bacteria with several benefits including protection of the bacterium from damage by encircling it with a tough, rigid structure. This structure is also porous. Small molecules are able to freely pass through the cell wall to the membrane, but large molecules are excluded. By performing this function, the cell wall acts as a coarse filter. The primary function of the cell wall, however, is to maintain the cell shape and prevent bursting due to osmotic pressure (called lysis).

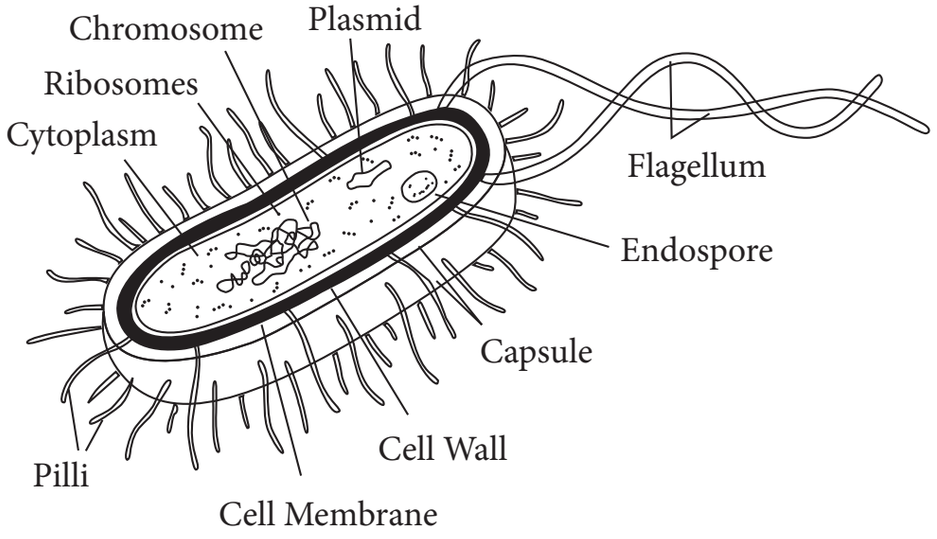


Figure 1.4 A schematic diagram of Bacteria

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